

CS-E4650 Methods of Data Mining, Homework N

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1 Methods

We used Python with NumPy for the computations and Matplotlib for the figures. The data file contains 1000 three-dimensional vectors with all values in $[0, 1]$ (checked in the code).

2 Histograms

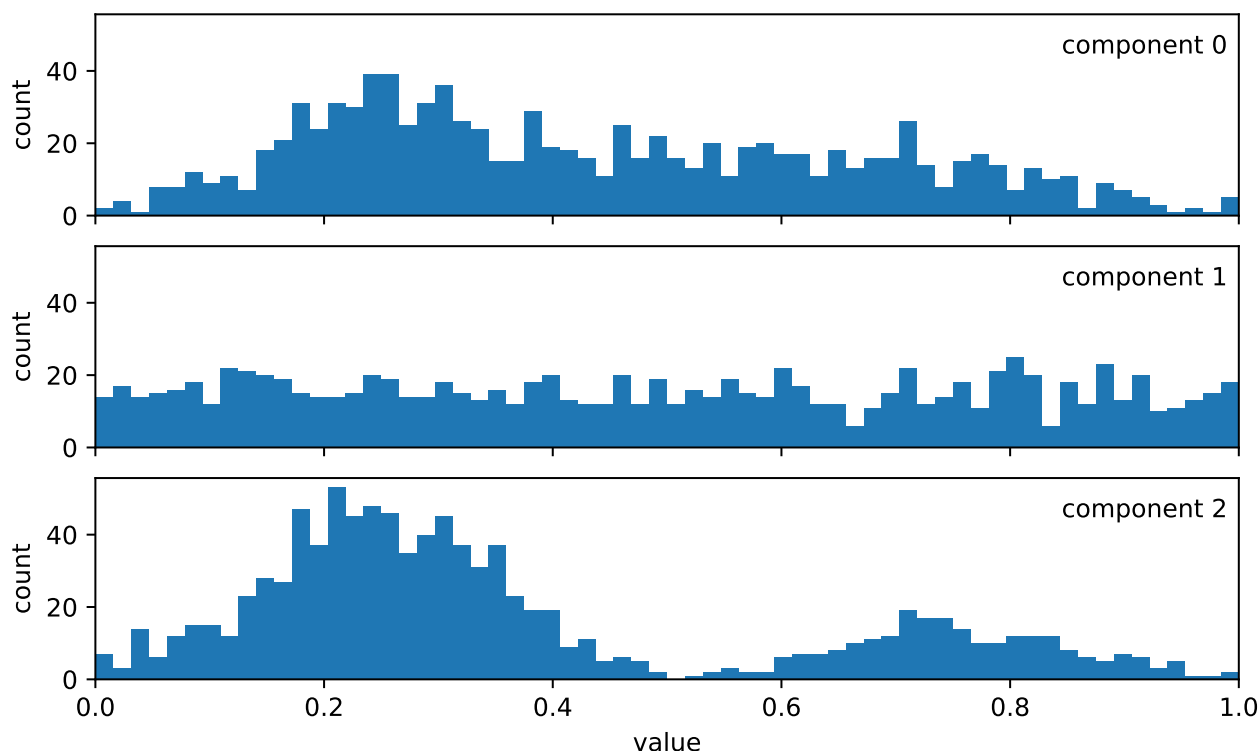


Figure 1: Histograms of the components (64 bins).

Component 0 (Figure 1) has a peak at about 0.25 and a broad shoulder up to about 0.9. It is clearly non-uniform and suggests overlapping groups, but there is no gap that would separate them, so on its own it shows only a weak clustering tendency. Component 1 is flat, with only random fluctuations, and shows no clustering tendency. Component 2 is clearly bimodal: about three quarters of the points lie around 0.25 and the rest around 0.75, separated by an almost empty gap at 0.5. It alone shows a strong clustering tendency with two groups.

3 Scatter plots

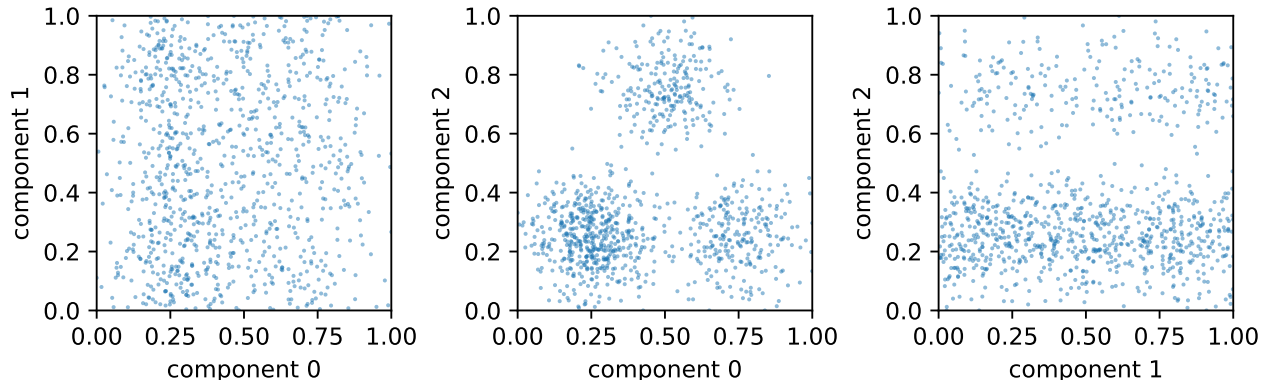


Figure 2: Scatter plots of the component pairs (equal aspect ratio).

Components 0 and 1 (Figure 2) show no groups: the density varies only along component 0, in the same way for every value of component 1, so the two components appear independent.

Components 0 and 2 show three compact, well-separated clusters. Their centres are at about $(0.25, 0.25)$, $(0.75, 0.25)$ and $(0.5, 0.75)$. The components are strongly but nonlinearly dependent: when component 2 is low, component 0 is around 0.25 or 0.75, and when component 2 is high, around 0.5. Neither component alone shows all three clusters: component 2 merges the two lower clusters, and in component 0 the upper cluster fills the gap between the lower two. The pair therefore has a clearly stronger clustering tendency than either component.

Components 1 and 2 show two horizontal bands (component 2 ≈ 0.25 and ≈ 0.75) spread evenly over component 1, i.e. only the bimodality of component 2 and no dependence on component 1.

Component 1 is thus uniform noise, independent of the other components. Including it stretches each cluster over the whole component-1 axis and dilutes the clustering tendency. The cluster structure lies in the subspace of components 0 and 2.

4 Entropies

Each subset of k dimensions was divided into $m = 64$ equal grid cells by splitting every dimension into $64^{1/k}$ equal ranges (4, 8 or 64 ranges for $k = 3, 2$ or 1 , with the value 1 in the last range). With p_i the fraction of points in cell i , the probability-based entropy of Aggarwal's eq. (6.2) is

$$E = - \sum_{i=1}^m [p_i \ln p_i + (1 - p_i) \ln(1 - p_i)]$$

with $0 \ln 0 = 0$. Since $m = 64$ for every subset, all values share the same range, from 0 (all points in one cell) to $E_{\max} = \ln 64 - 63 \ln \frac{63}{64} \approx 5.151$ (all $p_i = 1/64$).

All values in Table 1 lie between 4.482 and 5.118 (87 to 99 % of $E_{\max}$). $E_{\{0,2\}}$ is clearly the lowest, so $\{0, 2\}$ has the strongest clustering tendency. $E_{\{1\}}$ is within 1 % of $E_{\max}$ (practically uniform), and adding component 1 always increases the entropy (by 0.044 for $\{0\}$, 0.031 for

Table 1: Probability-based entropies of all combinations of dimensions ($m = 64$).

Dimensions	$\{0, 1, 2\}$	$\{0, 1\}$	$\{0, 2\}$	$\{1, 2\}$	$\{0\}$	$\{1\}$	$\{2\}$
E	4.794	4.998	4.482	4.792	4.954	5.118	4.761

$\{2\}$ and 0.312 for $\{0, 2\}$) because, with m fixed, an extra dimension lowers the resolution of the others without adding structure. Component 2 has the lowest single entropy (4.761), and adding it always decreases the entropy (by 0.472 for $\{0\}$, 0.326 for $\{1\}$ and 0.204 for $\{0, 1\}$), so it is the strongest single contributor. Component 0 alone is moderate (4.954) but lowers the entropy of $\{2\}$ by 0.279, so it contributes mainly together with component 2.

5 Greedy backward selection

Starting from all dimensions, each stage drops the worst feature, i.e. the one whose removal gives the lowest entropy, until every removal would increase the entropy (Table 2).

Table 2: Stages of greedy backward selection (lowest candidate in bold).

Stage	Current set (E)	Candidates (E)	Decision
1	$\{0, 1, 2\}$ (4.794)	$\{1, 2\}$: 4.792, $\{0, 2\}$: 4.482 , $\{0, 1\}$: 4.998	drop 1
2	$\{0, 2\}$ (4.482)	$\{2\}$: 4.761 , $\{0\}$: 4.954	stop

In stage 1, dropping component 1 gives by far the largest decrease (-0.312), whereas dropping component 0 barely changes the entropy (-0.002) and dropping component 2 increases it ($+0.204$). In stage 2, both removals increase the entropy ($+0.279$ and $+0.472$), so the search stops at $\{0, 2\}$ with $E = 4.482$.

6 Greedy forward selection

Starting from the empty set, each stage adds the best feature, i.e. the one that gives the lowest entropy, until every addition would increase the entropy (Table 3).

Table 3: Stages of greedy forward selection (lowest candidate in bold).

Stage	Current set (E)	Candidates (E)	Decision
1	$\emptyset$	$\{0\}$: 4.954, $\{1\}$: 5.118, $\{2\}$: 4.761	add 2
2	$\{2\}$ (4.761)	$\{0, 2\}$: 4.482 , $\{1, 2\}$: 4.792	add 0
3	$\{0, 2\}$ (4.482)	$\{0, 1, 2\}$: 4.794	stop

The empty set has no entropy on a 64-cell grid, so stage 1 takes component 2, the single dimension with the lowest entropy. In stage 2, adding component 0 decreases the entropy by 0.279, whereas adding component 1 increases it by 0.031. In stage 3, adding component 1 would increase the entropy by 0.312, so the search stops at $\{0, 2\}$, the same result as backward selection.

7 Analysis

The three approaches agree. Visually, the clusters lie in the plane of components 0 and 2 and component 1 is uniform noise. The entropies confirm this, and both greedy searches end at $\{0, 2\}$, which is also the minimum of all seven values and thus the result of an exhaustive search, as expected from the plots. The two searches agree because component 1 increases the entropy in every combination, so backward selection drops it first, while components 0 and 2 lower the entropy both alone and together, so forward selection adds them.

Two limitations are visible. First, with m fixed, the resolution per dimension falls with the dimensionality (64, 8 and 4 ranges). In the $4 \times 4 \times 4$ grid, the range boundaries 0.25, 0.5 and 0.75 cut through the cluster centres, so $E_{\{0,1,2\}}$ is not lower than $E_{\{1,2\}}$ (+0.002) although the full data contain all three clusters. Second, entropy measures non-uniformity rather than cluster separation. Component 0 has a clearly lower entropy than the uniform component 1 although its histogram shows no separated groups, so the visual inspection is needed to interpret the entropies.

8 Why 64 bins?

$64 = 4^3 = 8^2 = 64^1$, so uniform grids with equally many ranges per dimension give exactly $m = 64$ cells for every one-, two- and three-dimensional subset. Because the entropy depends on m , m should be the same for all compared subsets, which in general is only approximately possible with $\lceil m^{1/k} \rceil$ ranges per dimension. Equal m matters especially for the greedy searches, whose stopping decisions compare subsets of different dimensionality. In addition, $1000/64 \approx 15.6$ points per cell suffice to estimate the p_i .

Other values with the same property are the sixth powers 1, 64, 729, 4096, ... With $m = 1$ every entropy is 0, $m = 729 = 9^3 = 27^2$ leaves only 1.4 points per cell, and $m = 4096$ exceeds the number of points, so 64 is the only practical choice for these data.

With d -dimensional data, an equal m for all subset sizes $1, \dots, d$ requires $m = q^{\text{lcm}(1, \dots, d)}$, i.e. at least $2^{12} = 4096$ cells for $d = 4$ and 2^{60} for $d = 5$, which is impossible. Moreover, a k -dimensional grid with at least two ranges per dimension has at least 2^k cells, so the number of points needed grows exponentially and most cells become empty. In practice, m is therefore chosen from the number of points (e.g. at least 5 to 10 points per cell) with $\lceil m^{1/k} \rceil$ ranges per dimension, and entropies are compared relative to their maximum. Alternatively, the entropy can be computed from the one-dimensional distribution of pairwise distances, whose number of bins does not depend on the dimensionality.

9 Conclusions

The data have a clear clustering tendency only in the subspace of components 0 and 2, which contains three well-separated clusters. Component 1 is uniform, independent noise, component 2 is the most informative single component, and component 0 matters mainly together with component 2. The probability-based entropies and both greedy searches identify the same optimal subset, $\{0, 2\}$. Grid entropies are, however, sensitive to the grid resolution and measure non-uniformity rather than cluster separation, so they should be interpreted together with the visual inspection.

10 Appendix – Code

```
from itertools import combinations

import matplotlib.pyplot as plt
import numpy as np

X = np.loadtxt("data/clustering-tendency.csv", delimiter=",")
assert X.shape == (1000, 3) and X.min() >= 0 and X.max() <= 1

fig, axes = plt.subplots(3, 1, figsize=(7, 4.2), sharex=True, sharey=True,
                          layout="constrained")
for j, ax in enumerate(axes):
    ax.hist(X[:, j], bins=64, range=(0, 1))
    ax.text(0.99, 0.9, f"component {j}", transform=ax.transAxes,
            ha="right", va="top")
    ax.set(xlim=(0, 1), ylabel="count")
axes[2].set_xlabel("value")
fig.savefig("figures/histograms.pdf")

fig, axes = plt.subplots(1, 3, figsize=(7.5, 2.7), layout="constrained")
for ax, (i, j) in zip(axes, combinations(range(3), 2)):
    ax.scatter(X[:, i], X[:, j], s=3, alpha=0.5, linewidths=0)
    ax.set(xlim=(0, 1), ylim=(0, 1), aspect="equal",
           xlabel=f"component {i}", ylabel=f"component {j}")
fig.savefig("figures/scatter.pdf")

def entropy(dims, m=64):
    k = len(dims)
    counts, _ = np.histogramdd(X[:, dims], bins=round(m ** (1 / k)),
                               range=[[0, 1]] * k)
    p = counts.ravel() / len(X)
    q = np.concatenate([p, 1 - p])
    q = q[q > 0]
    return -np.sum(q * np.log(q))

for k in (3, 2, 1):
    for dims in combinations(range(3), k):
        print(dims, round(entropy(dims), 3))
```